

1. Problem Statement and Target Users

- What real-world problem does your application aim to solve?

The healthcare facilities such as hospitals and clinics are well known operating under enormous strain, driven by high patient demand as they are the primary points of care where people go for a wide range of health needs. Thus, staff may be busy and find it difficult to immediately identify urgent attention. This challenge is greater by the fact that many patients themselves may not fully realize the severity of their symptoms, leading them to seek routine care instead of urgent care. As a result, high patient volume and burnout for medical staff may cause potential delays in critical care for patients whose symptoms worsen while waiting.

There are some underlying symptoms (mild or severe) may or may not be happen immediately, the application is useful in ensuring patients who need urgent attention are spotted and treated more quickly, while the process for non-emergency care becomes more organized and efficient.

- Who are the intended users of the application?
 - Those who are hesitant to or are unsure about their symptoms warrant the need to see a doctor. The application would provide an initial assessment and help determine the next step for them.
 - Nurses to determine where they stand in the queue among other patients; they have so that the ones that urgently need it first can be prioritized.

2. User Inputs

- What information or data will users provide to the system?
 - Patient or Nurses
 - Symptoms
 - Symptoms history and time stamp
 - Name
 - NRIC (Nurses)
 - Age
 - Gender
 - Drug Allergy

3. Use of AI

- How will AI be utilized within the application?
 - AI will be used to process information given by the user and determine what to suggest to the user on what to do next.

- It will ask follow-up questions and such to gain more information before deciding
- It will give the severity of symptoms whether it requires medical care
- What outputs, insights, or recommendations will the AI generate from the user inputs?
 - The AI will take the symptoms and form an initial diagnosis and recommend the appropriate next step for them, whether it is serious enough to go see a doctor or to wait out and monitor.
 - The AI will keep a log which the user can print or email the log to send to the healthcare provider to aid them better.

4. Business Rules

- What business rules, validations, or decision-making logic will be applied to the AI-generated outputs?
 - AI Failure handling
 - If the AI service fails, it will display an error message and ask the user to reload page or wait for a few minutes if the error persists.
 - Emergency symptom detection
 - difficulty breathing and severe bleeding. It will prioritize recommending getting emergency medical aid.
 - Severity weightage
 - Different levels of urgency like urgent or un-urgent
 - Contradictory or incomplete information.
 - Will prompt user for more information if information given is unclear or incomplete
 - Uncertainty and confidence handling
 - Will ask follow-up questions instead of making assumptions
 - Will recommend a consultation if it cannot determine condition
 - Able to filter important information can be displayed for user to take note of
 - History of how the user get the injury or symptom
 - History of what solutions the AI can give
 - Recommendations given to user
 - If it is an injury eg a cut, the application will give steps how to remediate it instead of letting the user figuring it out by himself